

Rupam Biswas

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Summary

Master's student in Data Science at TU Dortmund University with professional experience in software engineering, machine learning, artificial intelligence, and backend development. Experienced in Python, .NET Core, SQL Server, deep learning, and real-time data systems. Co-author of research work in deep learning and federated learning for medical image analysis. Passionate about building intelligent systems and applying data-driven solutions to real-world challenges. Currently seeking Werkstudent opportunities in Software Engineering, Data Science, Artificial Intelligence, or Machine Learning.

Education

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| TU Dortmund University , Master of Science (M.Sc.) in Data Science | April 2026 – Present |
| • Currently pursuing M.Sc. in Data Science | |
| Gujarat Technological University , Bachelor of Engineering in Computer Engineering (Govt. ICCR Scholarship) | Sept 2020 – June 2024 |
| • Final Grade (German Equivalent): 1.5/5 | |
| Gujarat Technological University , Specialized Minor Degree in IoT | Jan 2022 – Dec 2023 |
| • Implementation of AI techniques in IoT systems | |

Experience

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|---|---------------------|
| Associate Software Engineer , Data Edge Limited | Feb 2025 – Apr 2026 |
| • Developed a real-time transaction monitoring system for Trust Bank PLC using Python and .NET Core, integrating a Linux-based backend with a Windows-based frontend and SQL Server database. | |
| • Contributed to an ATM reconciliation system for BRAC Bank PLC using C#, .NET, and SQL in a production environment. | |
| Data Science Intern , SAP India – CodeUnnati | Jan 2024 – Apr 2024 |
| • Worked on projects involving machine learning, image processing, and IoT integration. | |

Projects

Real-Time Bank Transaction Monitoring System

- Designed and developed a real-time transaction monitoring platform for Trust Bank PLC.
- Built a Linux-based backend using Python and integrated it with a .NET Core frontend and SQL Server database.
- Implemented live transaction processing, monitoring, filtering, and operational dashboards.
- Technologies: Python, .NET Core, SQL Server, SignalR, Linux
- Successfully deployed and currently used in a production banking environment.

Plant Disease Identification and Solution System

[GitHub Link](#)

- Built a deep learning model to identify plant diseases from leaf images, allowing users to upload or capture photos instantly. Integrated a solution recommendation system using Meta LLaMA LLM. Deployed via Streamlit.
- Tools used: Python, Streamlit, TensorFlow, Keras, LLaMA (LLM)

Multilingual-Retrieval-Augmented-Generation-RAG-System

[GitHub Link](#)

- Developed a multilingual Retrieval-Augmented Generation (RAG) system capable of answering both English and Bengali queries using document-based knowledge retrieval.
- Built an end-to-end pipeline incorporating OCR, document chunking, vector embeddings, FAISS vector search, conversational memory, and OpenAI-powered response generation.

- Implemented REST APIs using FastAPI and designed support for both long-term document memory and short-term conversational memory.
- Technologies used: Python, FastAPI, OpenAI API, FAISS, Tesseract OCR, RAG, Vector Databases, NLP.

GenAI-based Local Search Engine

[GitHub Link](#)

- Created a search engine that recommends learning resources based on user interests using embeddings and attention mechanisms. Fine-tuned a LLM model from Hugging Face for semantic understanding.
- Tools used: Python, Hugging Face, Transformers, Attention, LLMs

Publications & Research

LungNet-CAM: A Hybrid Convolution and Attention-Based Deep Learning Framework for CT-Based Lung Cancer Detection

- Co-author
- Accepted and presented at [IEEE QPAIN], 2026

Improving Non-IID Based Tumor Classification Through Resource-Efficient Federated Learning with Attention Ensemble CNNs

- Co-author
- Submitted for peer at [SPICSCON2026] IEEE international conference

Technologies

Languages: Python, C#, SQL

Frameworks & Tools: . NET Core, ASP.NET Core, FastAPI, Flask, Django, SignalR, PowerBI, Tableau, Git, Docker

Data Science AI: TensorFlow, Keras, Scikit-learn, Hugging Face, Transformers, FAISS, OpenAI API, RAG

Areas of Expertise: Machine Learning, Deep Learning, Large Language Models (LLMs), Data Analysis, Backend Development, AI Agents, Business Intelligence, Real-Time Data Systems

Communication And Availability

- **English:** Fluent (Professional)
- **Bengali:** Native
- **German:** A1 Basic (Actively improving and attending language courses)
- Valid student work permit in Germany
- Available for Werkstudent positions
- Flexible working hours including weekends and holidays